

Lecture 14: Floating Point Arithmetic

CMPS 221 – Computer Organization and Design

Floating-Point Addition

- Consider a 4-digit decimal example
 - $9.999 \times 10^1 + 1.610 \times 10^{-1}$
- 1. Align decimal points
 - Shift number with smaller exponent
- 2. Add significands
- 3. Normalize result & check for over/underflow
- 4. Round and renormalize if necessary

$$\begin{array}{r} 9.999 \times 10^1 \\ + 1.610 \times 10^{-1} \\ \hline \end{array}$$

$$\begin{array}{rcl} & & 9.999 \times 10^1 \\ \text{Align:} & + & 0.016 \times 10^1 \\ \hline & & 10.015 \times 10^1 \\ \text{Add:} & & \\ \text{Normalize:} & & 1.0015 \times 10^2 \\ \text{Round:} & & 1.002 \times 10^2 \end{array}$$

Floating-Point Addition

- Now consider a 4-digit binary example
 - $1.111_2 \times 2^{-3} + 1.100_2 \times 2^{-5}$
- 1. Align binary points
 - Shift number with smaller exponent
- 2. Add significands
- 3. Normalize result & check for over/underflow
- 4. Round and renormalize if necessary

~~$$\begin{array}{r}
 1.111 \times 2^{-3} \\
 + 1.100 \times 2^{-5} \\
 \hline
 \end{array}$$~~

Align:

$$\begin{array}{r}
 1.111 \times 2^{-3} \\
 + 0.011 \times 2^{-3} \\
 \hline
 \end{array}$$

Add:

$$10.010 \times 2^{-3}$$

Normalize:

$$1.0010 \times 2^{-2}$$

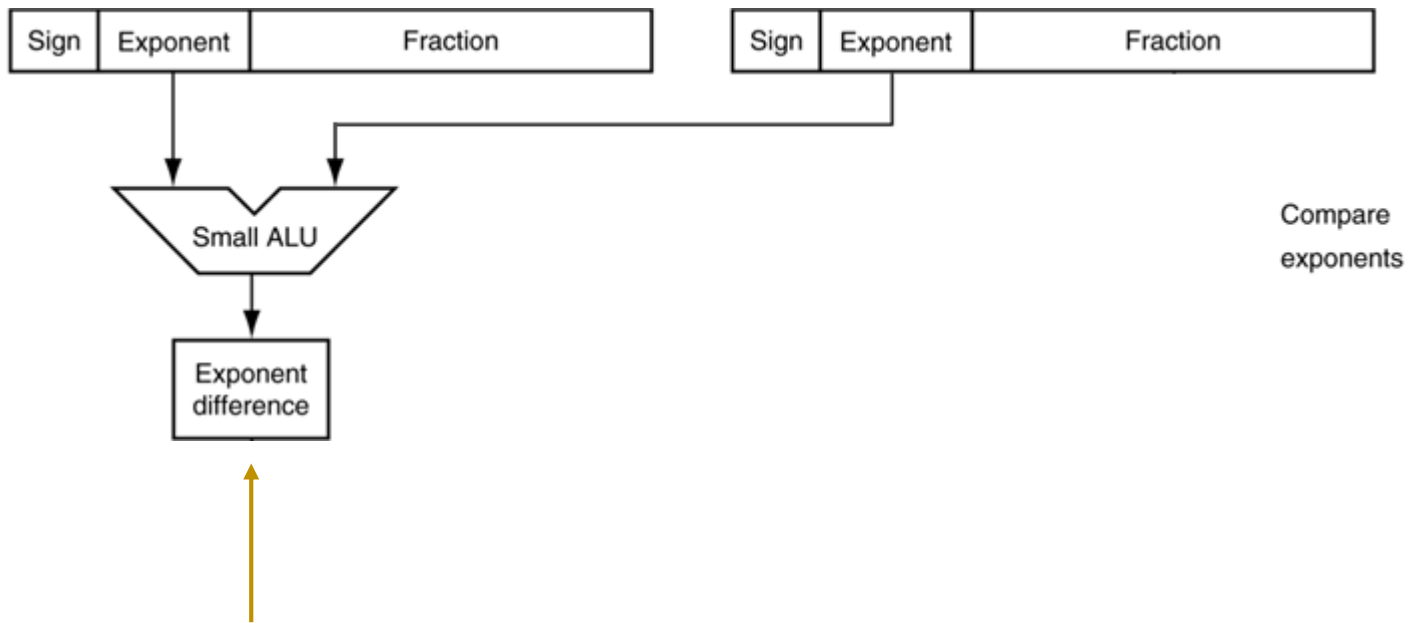
Round:

$$1.001 \times 2^{-2}$$

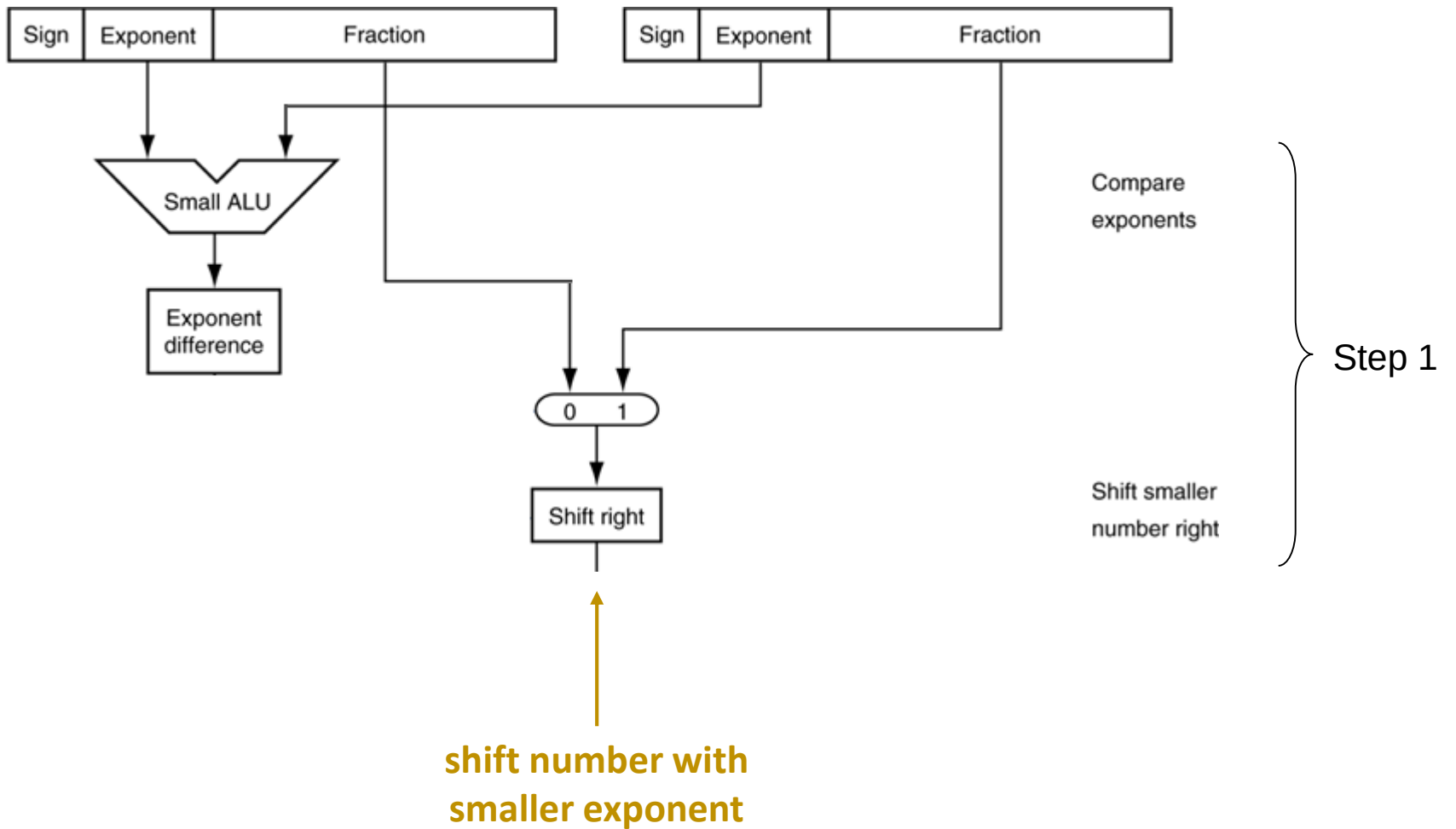
*How to do this in
hardware?*

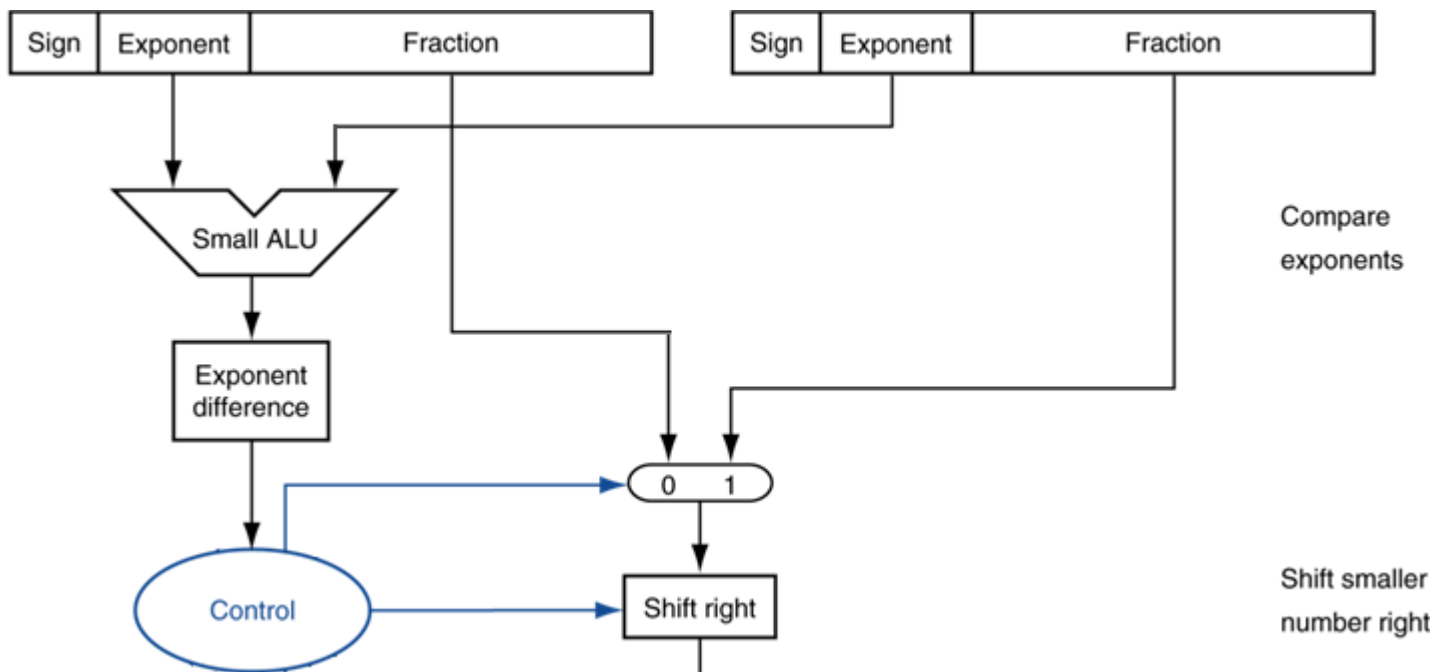
Sign	Exponent	Fraction
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Sign	Exponent	Fraction
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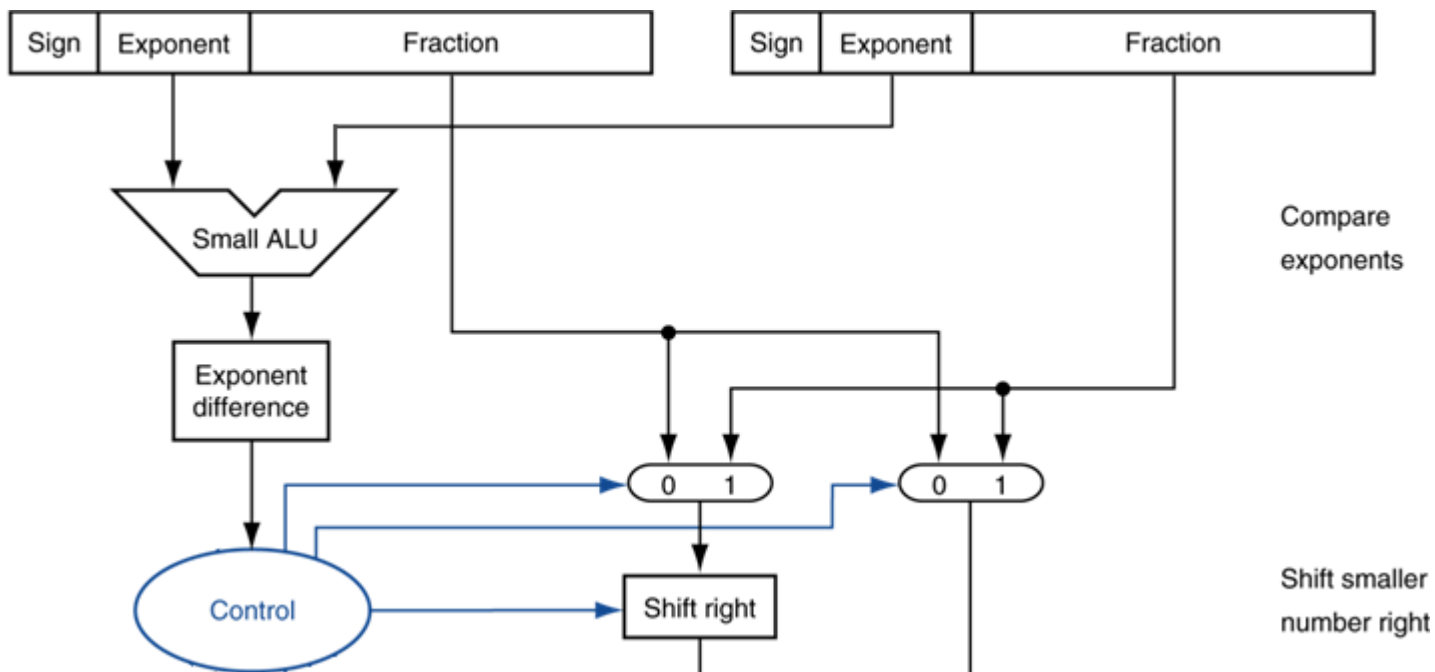
**subtract exponents to find
which number is smaller**





use exponent difference to determine which value to shift and by how much

shift number with smaller exponent

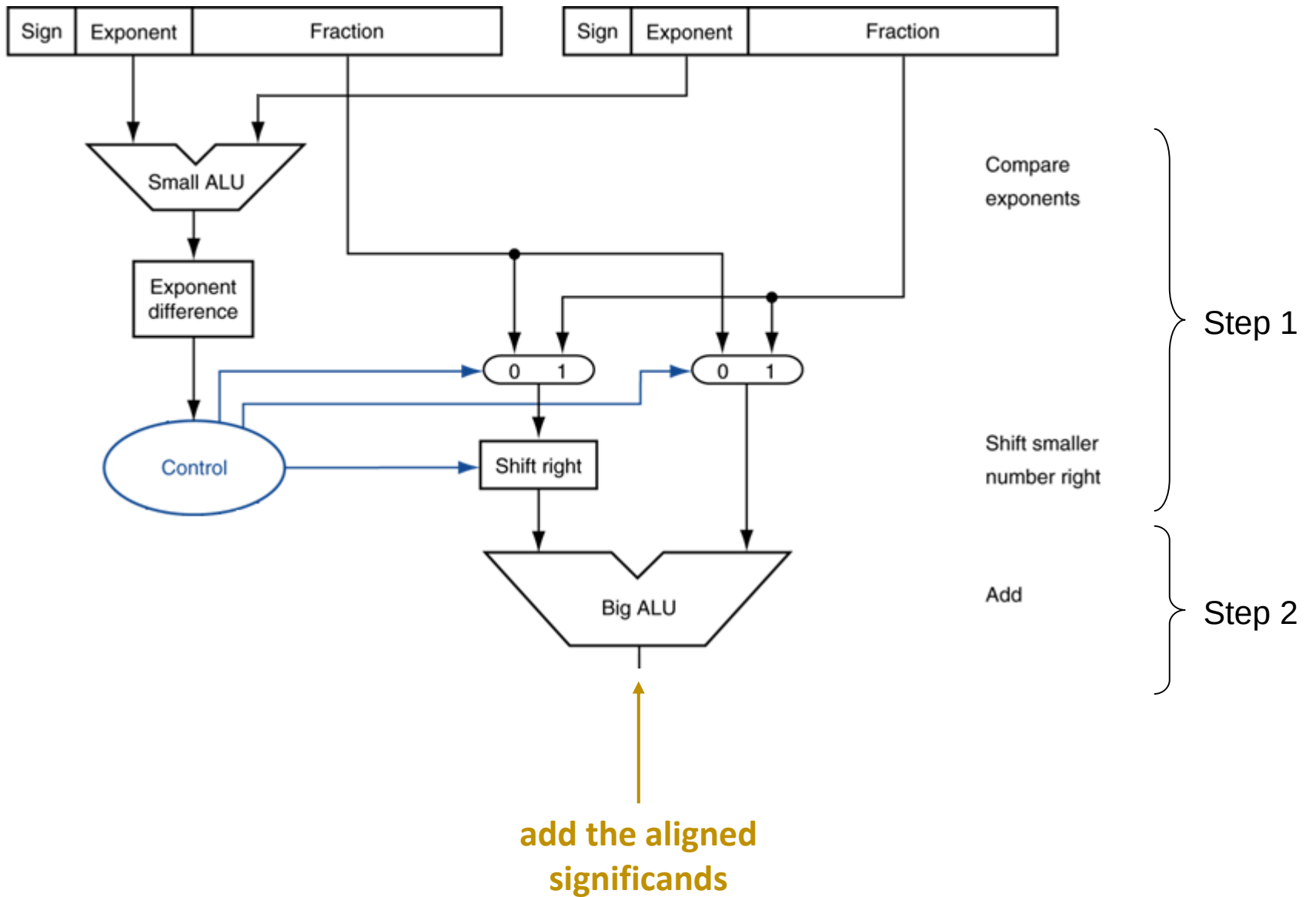


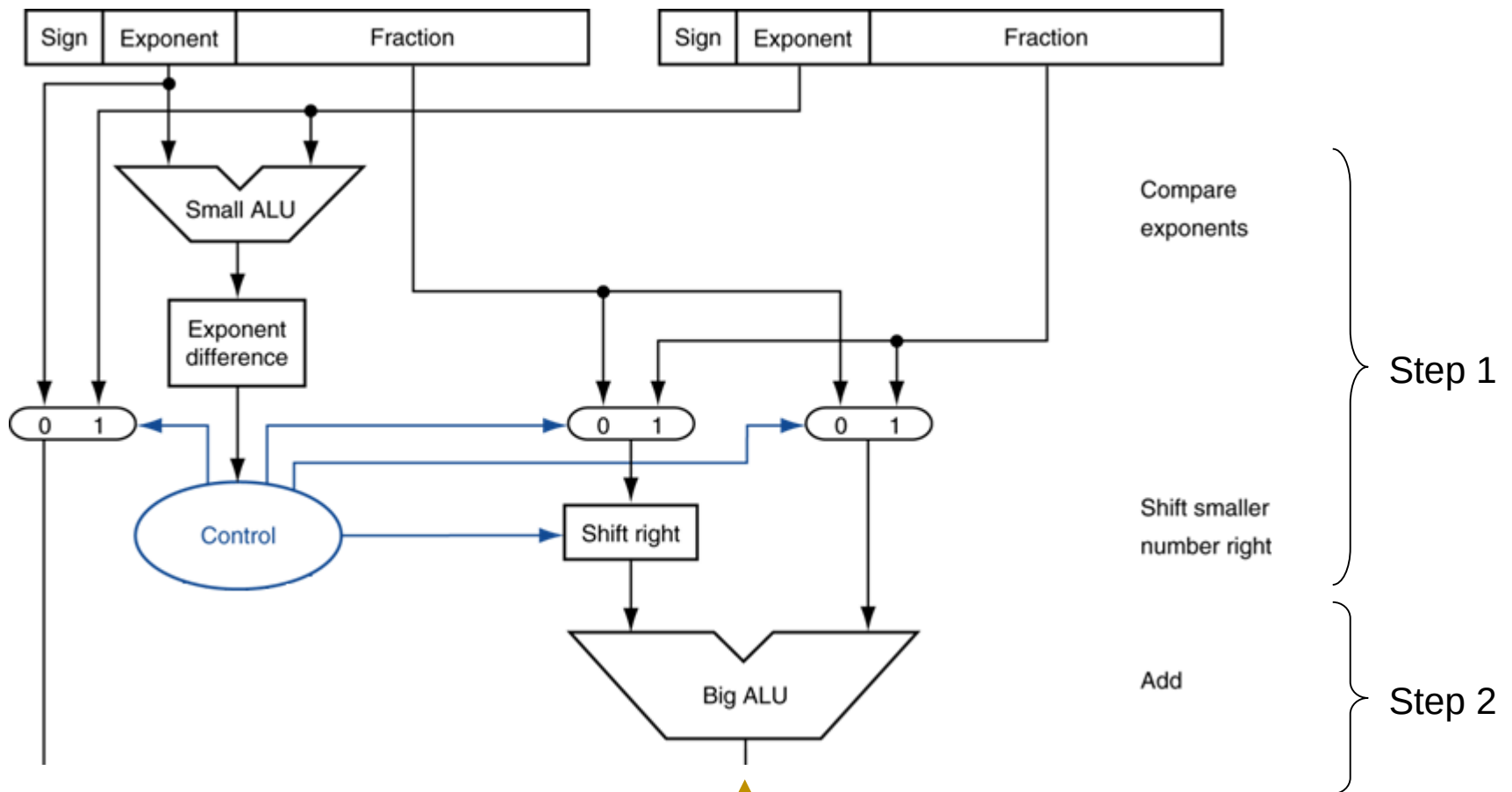
Step 1

use exponent difference to determine which value to shift and by how much

shift number with smaller exponent

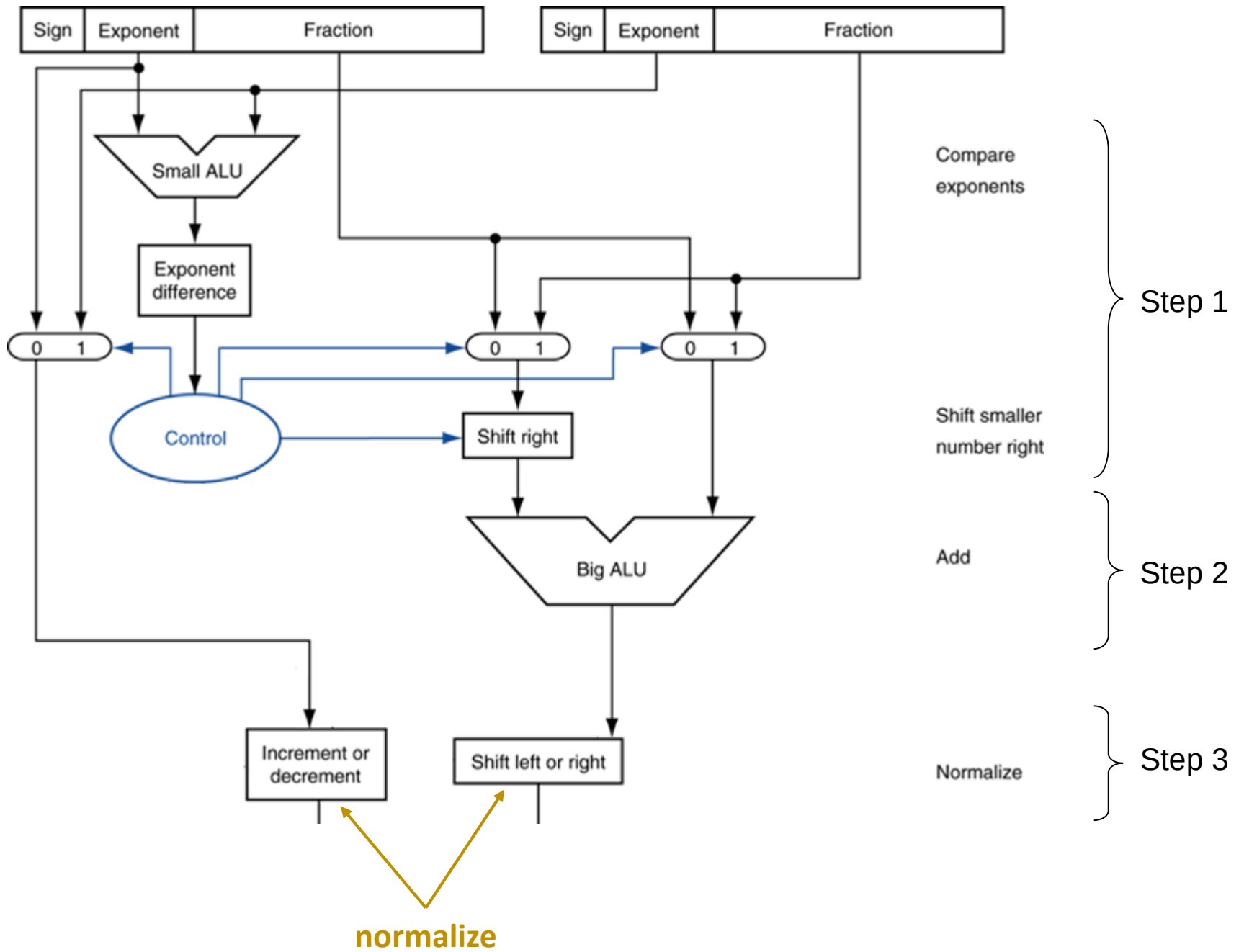
keep number with larger exponent as is

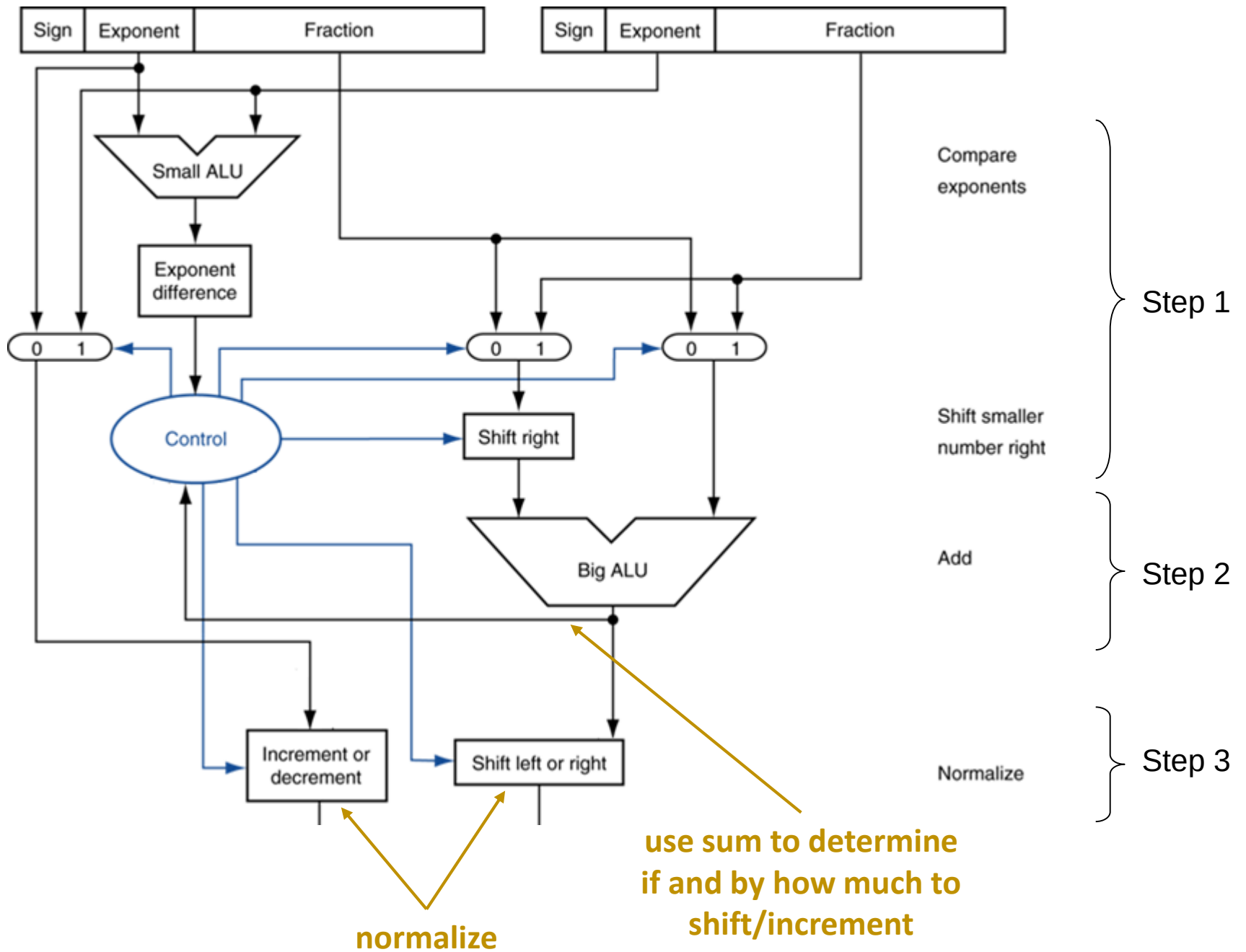


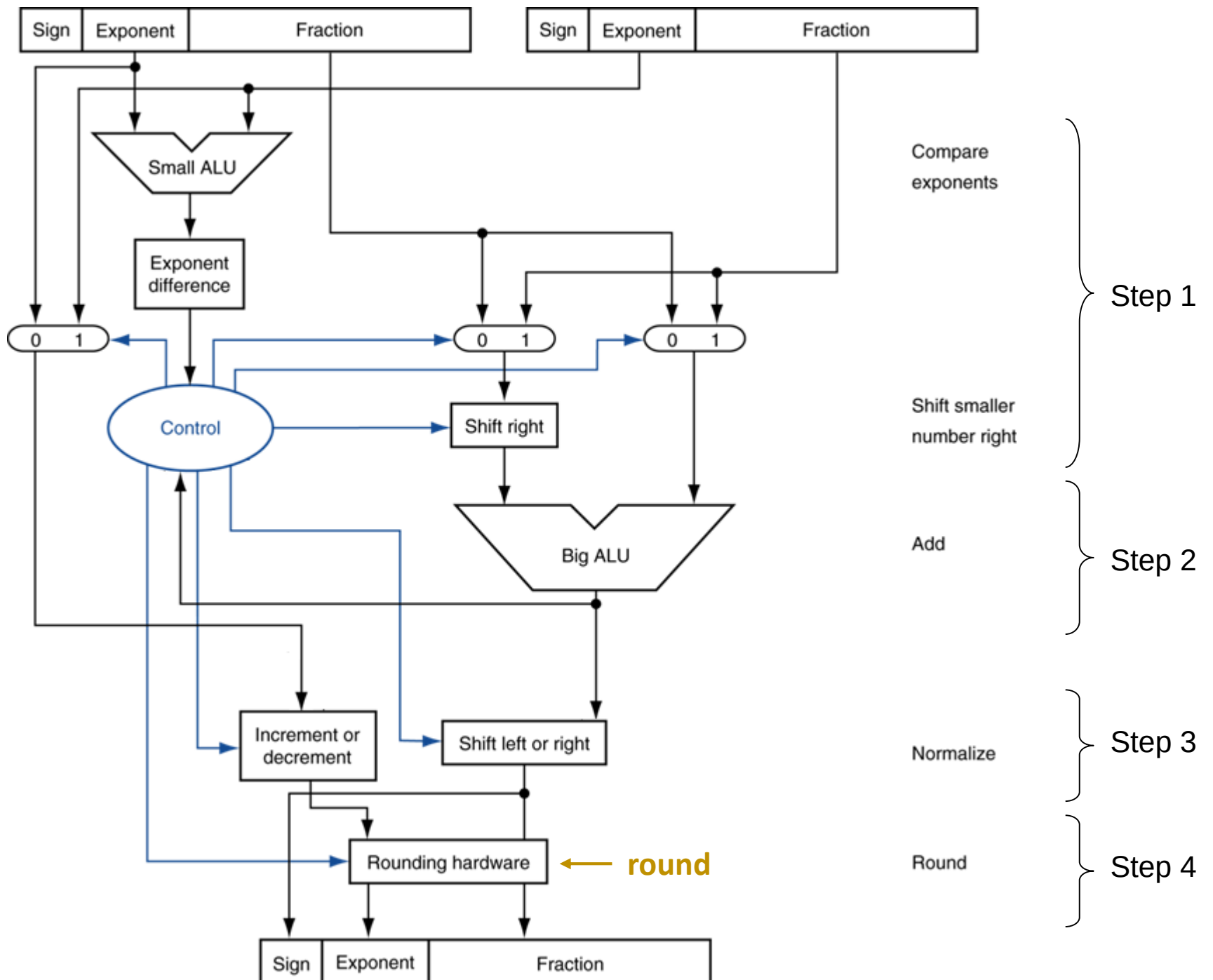


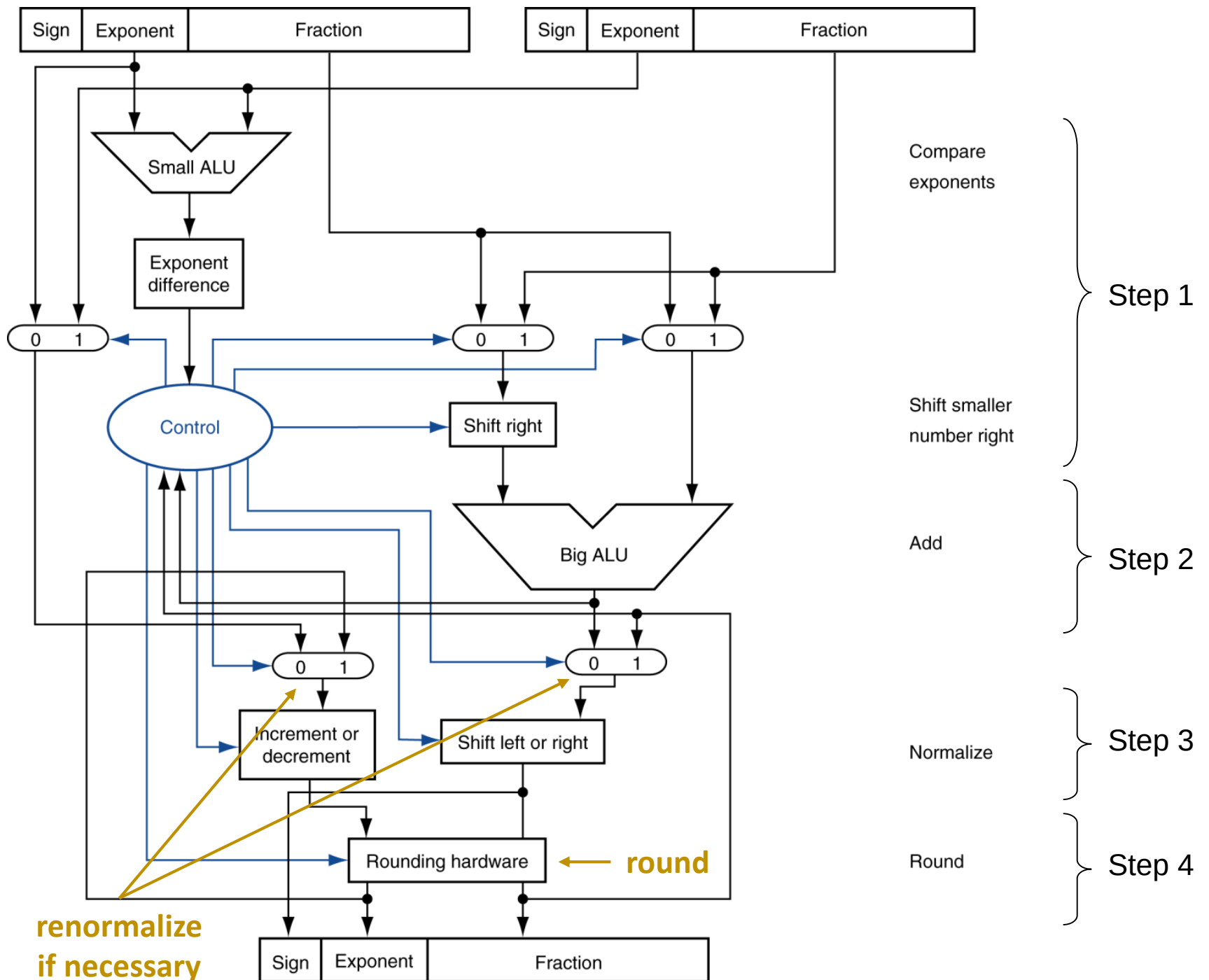
choose the larger
exponent as the
exponent of the sum

add the aligned
significands









FP Adder Hardware

- Much more complex than integer adder
- Doing it in one cycle would take too long
 - Much longer than integer operations
 - Slower clock would penalize all instructions
- FP adder usually takes several cycles
 - Can be pipelined

Floating-Point Multiplication

- Consider a 4-digit decimal example
 - $4.230 \times 10^{10} \times 9.200 \times 10^{-5}$
- 1. Add exponents
 - For biased exponents, subtract bias from sum
 - New exponent = $10 + -5 = 5$
- 2. Multiply significands
 - $4.230 \times 9.200 = 38.916 \Rightarrow 38.916 \times 10^5$
- 3. Normalize result & check for over/underflow
 - 3.8916×10^6
- 4. Round and renormalize if necessary
 - 3.892×10^6
- 5. Determine sign of result from signs of operands
 - $+3.892 \times 10^6$

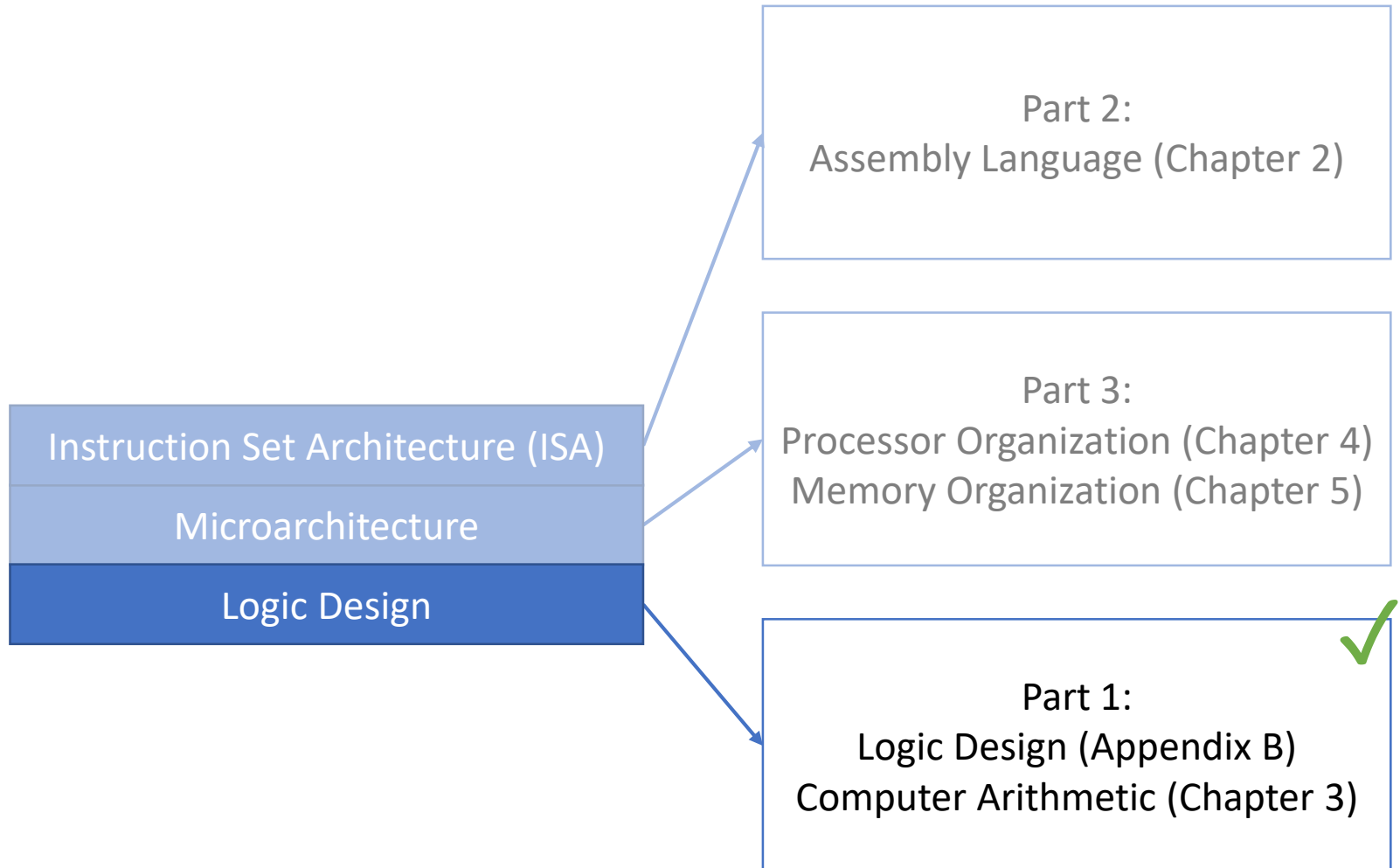
Floating-Point Multiplication

- Now consider a 4-digit binary example
 - $1.010_2 \times 2^2 \times -1.101_2 \times 2^{-7}$
- 1. Add exponents
 - $2 + -7 = -5$
- 2. Multiply significands
 - $1.010_2 \times 1.101_2 = 10.000010_2 \Rightarrow 10.000010_2 \times 2^{-5}$
- 3. Normalize result & check for over/underflow
 - $1.0000010_2 \times 2^{-4}$
- 4. Round and renormalize if necessary
 - $1.000_2 \times 2^{-4}$
- 5. Determine sign: +ve $\times$ -ve $\Rightarrow$ -ve
 - $-1.000_2 \times 2^{-4}$

FP Arithmetic Hardware

- FP multiplier has similar complexity to FP adder
 - But uses a multiplier for significands instead of an adder
- FP arithmetic hardware usually does
 - Addition, subtraction, multiplication, division, reciprocal, square-root
 - $\text{FP} \leftrightarrow \text{integer}$ conversion
- Operations usually takes several cycles
 - Can be pipelined

End of Part 1



Textbook Sections

- The content in these slides corresponds to:
 - Textbook:
 - *Computer Organization and Design, 5th Edition by David Patterson and John Hennessy, Morgan Kaufmann, 2014.*
 - Sections:
 - 3.5